

RStudio, Git, and GitHub Configuration Tutorial

A step by step guide for a first time user

RISE Data Science Team

22-07-2024

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For those who have never used GitHub before, it can be very intimidating. But don't worry! We created this tutorial to help you understand how to configure Git in your computer, open a GitHub account, and to link your RStudio projects with your GitHub repositories. This tutorial is designed for someone who has little/no experience with R or RStudio, and no experience with GitHub. Please be patient going through this tutorial and do not hesitate to reach out if you have questions.

0.1 Step 1: Download Git

First, you need to install Git, a distributed control system designed to handle everything from small to very large projects with speed and efficiency.

To install Git, go to the Git website [here](https://git-scm.com/).

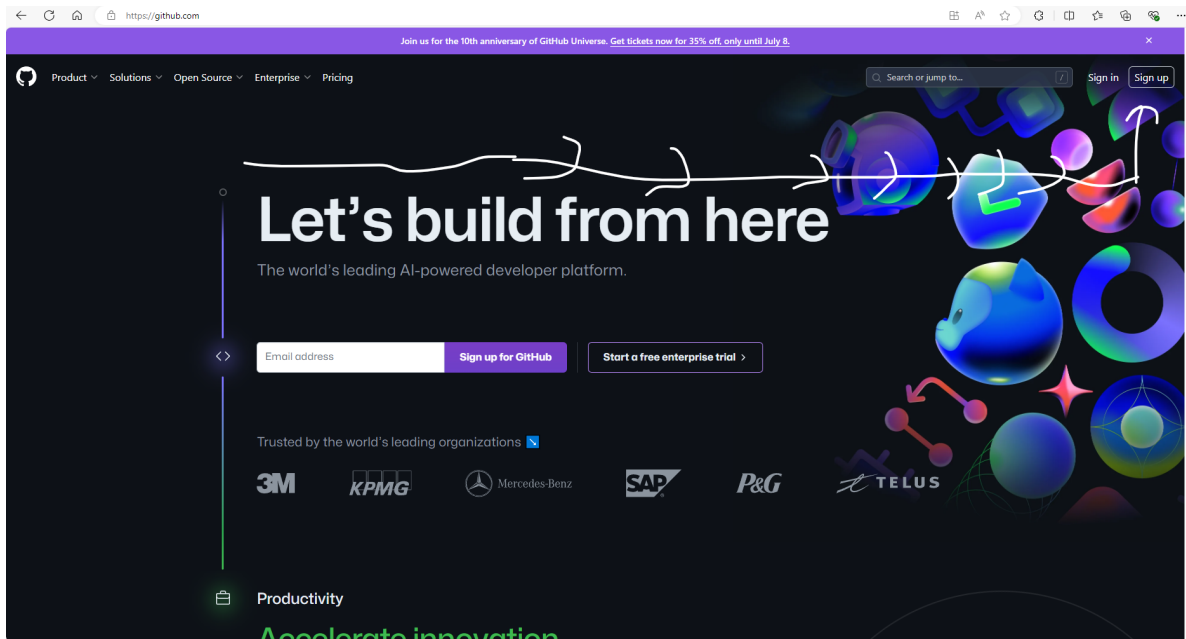
Then, click download for your operative system (e.g., Windows) and allow it to install. Accept the standard recommendations during the installation.

0.2 Step 2: Create a Github account

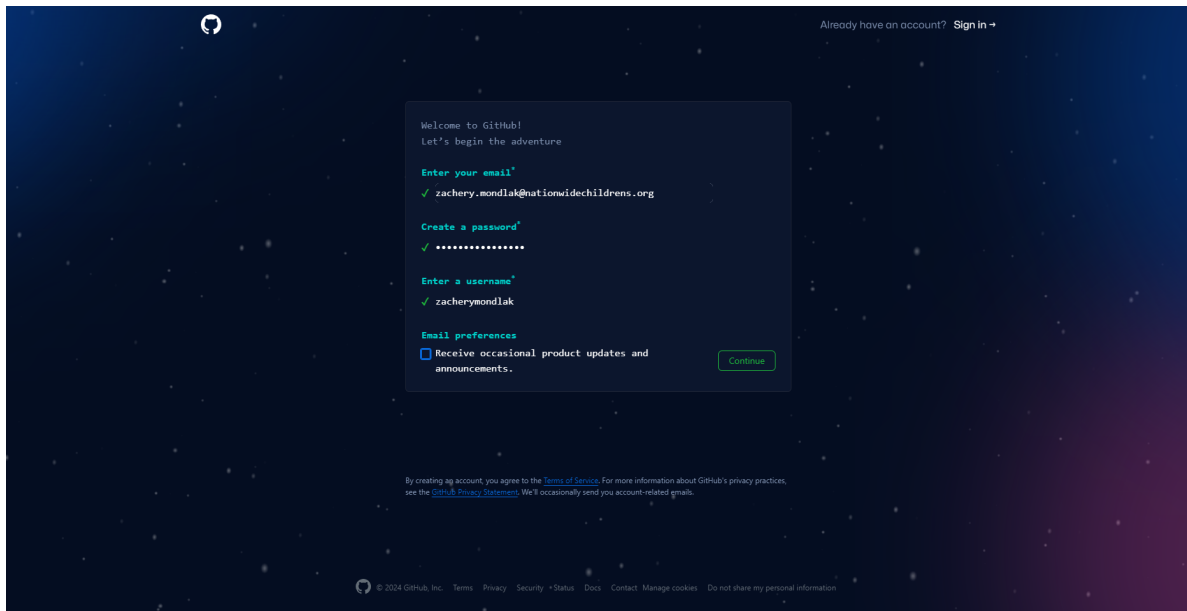
Now, let's create a GitHub account. GitHub is a web-based platform that uses Git for version control and source code management. It provides a collaborative environment for teams to work on projects together.

Let's go the GitHub page [here](https://github.com).

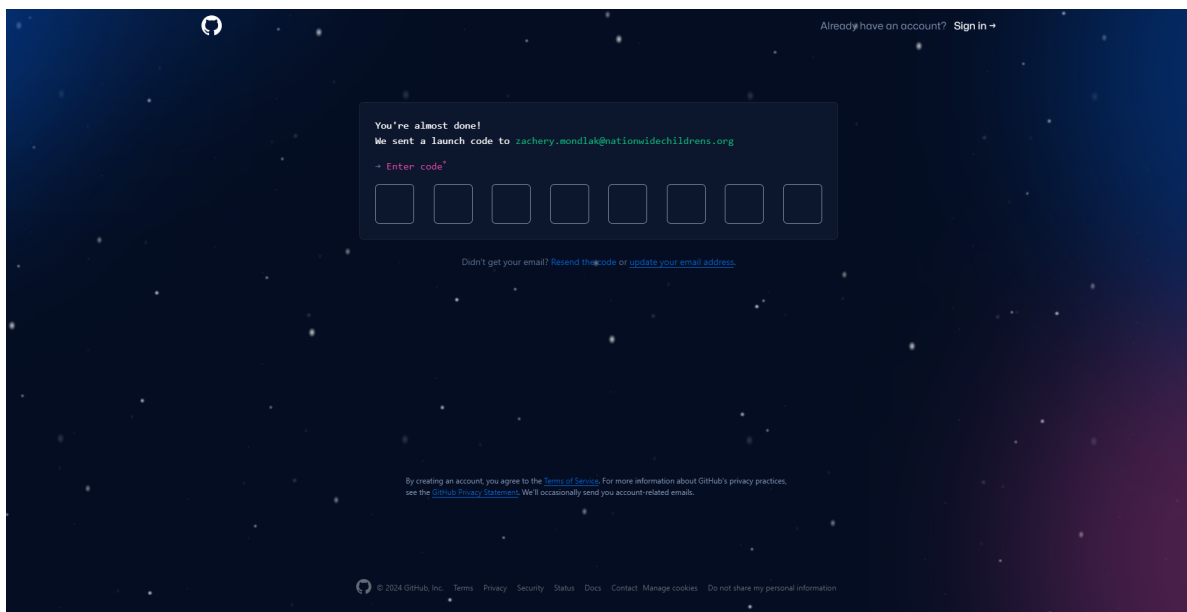
Inside the welcome page, on the top right, click Sign Up to create a new account.



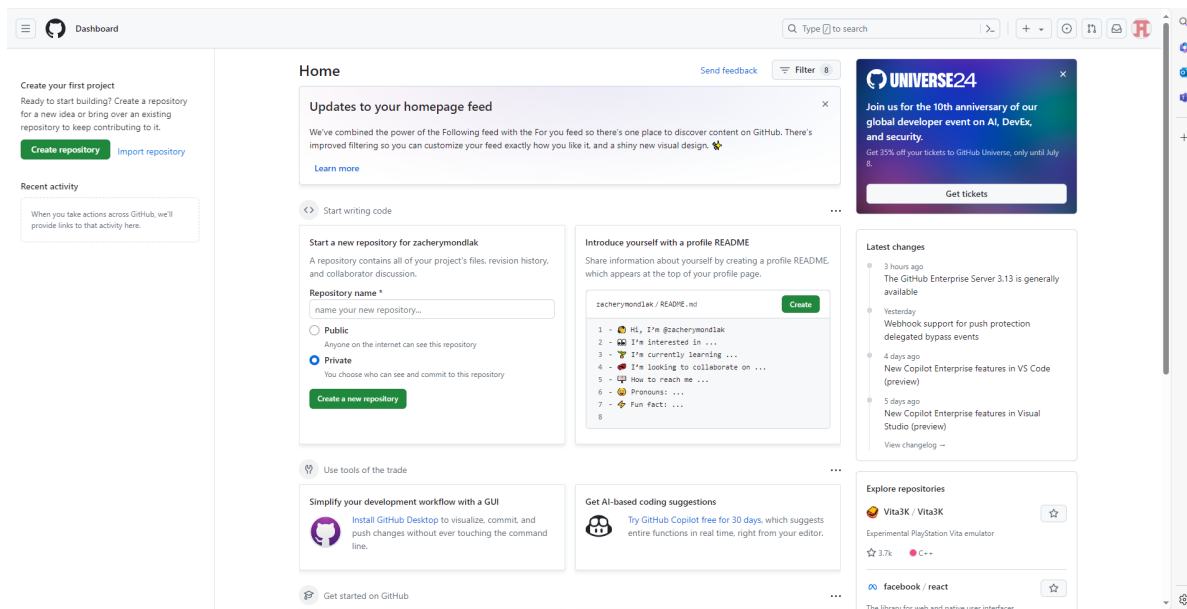
Use your preferred email and password (Note, it is only Nationwide for demo purposes)



Verify your account through both security procedures (picture, then code)



After verifying, you will see the following page:



0.3 Step 3: Setting up Git in RStudio to connect with GitHub

To do the setup, we are using the R Package “usethis”

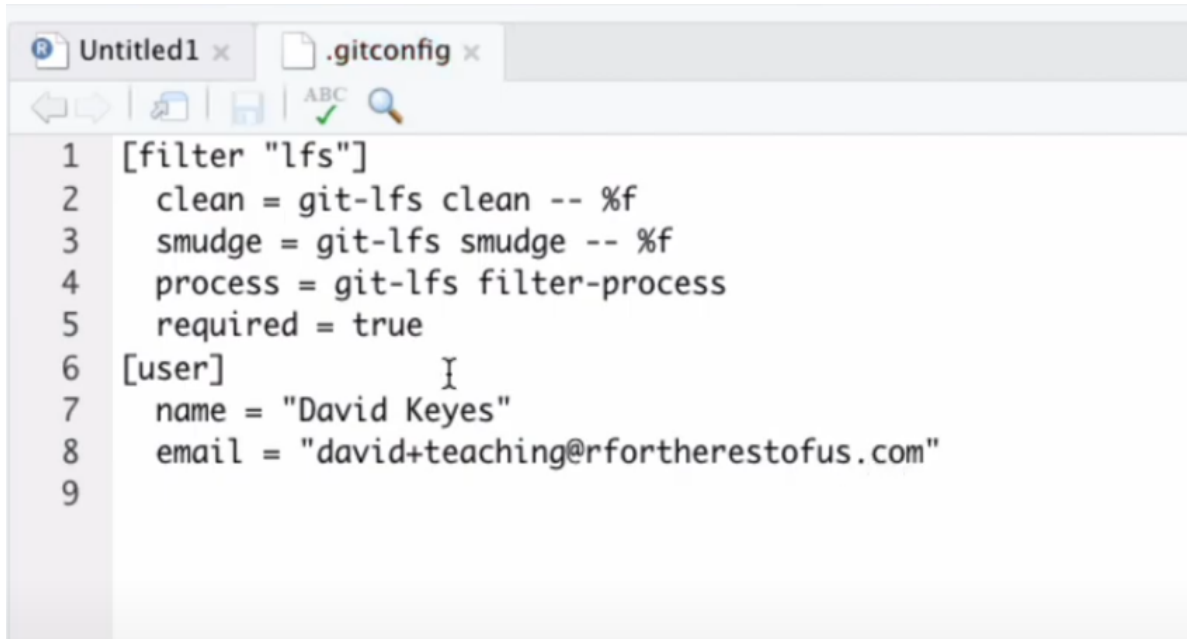
Let’s start running the code below.

```
install.packages ("usethis")
library (usethis)

edit_git_config()
```

This should should open a .gitconfig tab in R as in the figure below.

There, you enter your name and email address (lines 7 and 8 in example) and click enter. Use the same credentials used to create your GitHub account.

A screenshot of an RStudio editor window. The top bar shows two tabs: 'Untitled1' and '.gitconfig'. The '.gitconfig' tab is active. The editor displays the following configuration for Git LFS:

```
1 [filter "lfs"]
2   clean = git-lfs clean -- %f
3   smudge = git-lfs smudge -- %f
4   process = git-lfs filter-process
5   required = true
6 [user]
7   name = "David Keyes"
8   email = "david+teaching@rfortherestofus.com"
9
```

0.4 Step 3.5: Create a personal access token to connect RStudio to GitHub

Now, you have one more thing to do.

Run this code to create a personal access token. A personal access token in GitHub is a secure, alphanumeric string that grants access to your GitHub account and can be used instead of a password for authentication.

```
usethis::create_github_token()
```

It should take you to a new web page outside RStudio that looks like this:

Settings / Developer settings

GitHub Apps
OAuth Apps
Personal access tokens

New personal access token

Personal access tokens function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

Note

What's this token for?

Select scopes

Scopes define the access for personal tokens. [Read more about OAuth scopes.](#)

☒ repo	Full control of private repositories
☒ repo:status	Access commit status
☒ repo_deployment	Access deployment status
☒ public_repo	Access public repositories
☒ repo:invite	Access repository invitations
☒ security_events	Read and write security events
☒ workflow	Update GitHub Action workflows
☐ write:packages	Upload packages to GitHub Package Registry
☐ read:packages	Download packages from GitHub Package Registry
☐ delete:packages	Delete packages from GitHub Package Registry
☐ admin:org	Full control of orgs and teams, read and write org projects
☐ write:org	Read and write org and team membership, read and write org projects
☐ read:org	Read org and team membership, read org projects

Give your personal access token a name (whatever you want!).

Then scroll to the bottom of the page and click ‘Generate new token’. It will take you to a new page where you can copy your personal access token:

Settings / Developer settings

GitHub Apps
OAuth Apps
Personal access tokens

Personal access tokens

Generate new token Revoke all

Tokens you have generated that can be used to access the GitHub API.

Make sure to copy your new personal access token now. You won't be able to see it again!

☒ 2f6036ef59958c4710ce8475ed72913b92c4f1cf
Delete

Personal access tokens function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

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Copy your personal access token then go back to the console in Rstudio. It is recommend to save a copy of the taken in a secure file in your computer.

Type use this code that will open a dialog asking you to enter your token.

```
gitcreds::gitcreds_set()
```

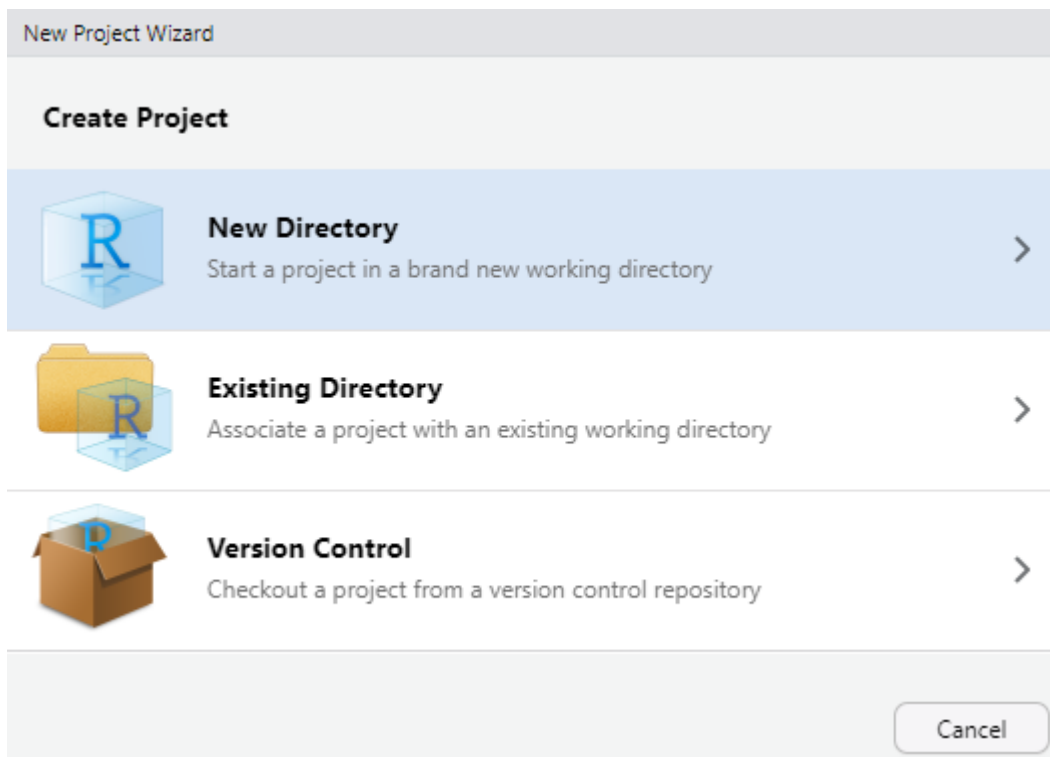
Good work! You have now configured Rstudio to be linked to your Github account!

0.5 Step 4: Sharing your local RStudio projects

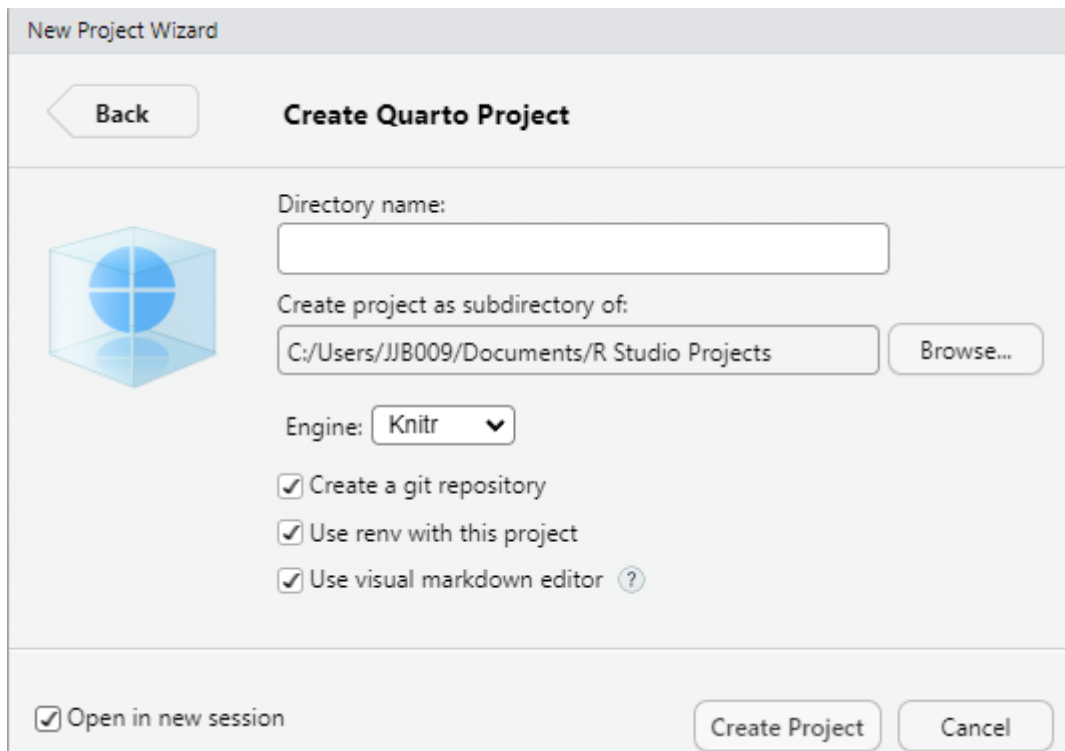
Now, you can share your RStudio projects with your team, or use GitHub as a backup for your projects.

Let's create new project.

First, in the top left corner of RStudio, click File, then click new project, and it should take you to this page:



For this example, click on New Directory and choose a project type (typically a Quarto project for the data science team). It will take you to this page:



New Project Wizard

Back **Create Quarto Project**

 Directory name:

Create project as subdirectory of: Browse...

Engine:

☒ Create a git repository

☒ Use renv with this project

☒ Use visual markdown editor ?

☒ Open in new session

Create Project Cancel

Choose the name of the new RStudio project you are creating (also make sure to check all boxes) and click create new project. This should take you to a new RStudio session.

In the console area, run this code:

```
usethis::use_git()
```

When RStudio asks you questions in the console, select yes for all options (they may have it worded “definitely” or “For sure”).


```

> library(usethis)
> use_git()
✓ Setting active project to '/Users/teacher/Desktop/testrepo'
✓ Initialising Git repo
✓ Adding '.Rproj.user', '.Rhistory', '.Rdata', '.httr-oauth', '.DS_Store' to '.gitignore'
There are 2 uncommitted files:
* '.gitignore'
* 'testrepo.Rproj'
Is it ok to commit them?

1: Definitely
2: Not now
3: Negative

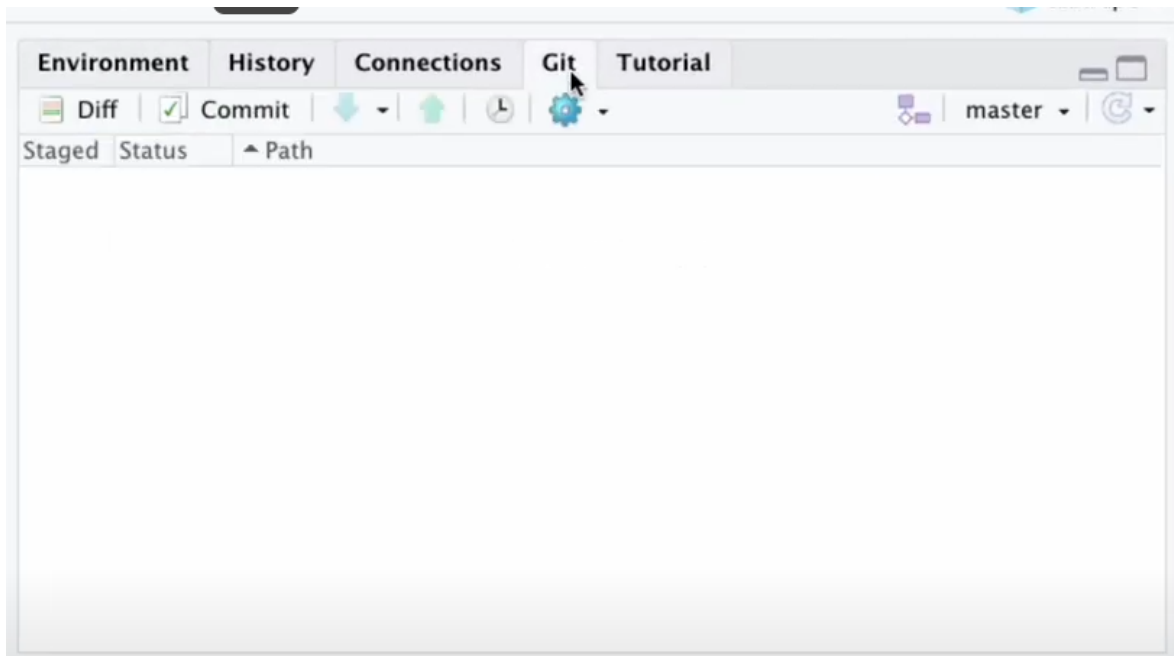
Selection: 1
✓ Adding files
✓ Making a commit with message 'Initial commit'
• A restart of RStudio is required to activate the Git pane
Restart now?

1: For sure
2: Absolutely not
3: No way

Selection: 1

```

Now, you should have initialized your new project as a Git repository. Check to see if you have the Git tab in the top right section of R studio:



Phew! You've gotten a lot accomplished so far! You now have created your first local Git

repository.

There is plenty of more to learn, but for now, let's stop here.

0.6 Step 4.5: Sharing your local RStudio project in GitHub

here, we need to add `usethis::use_github()`

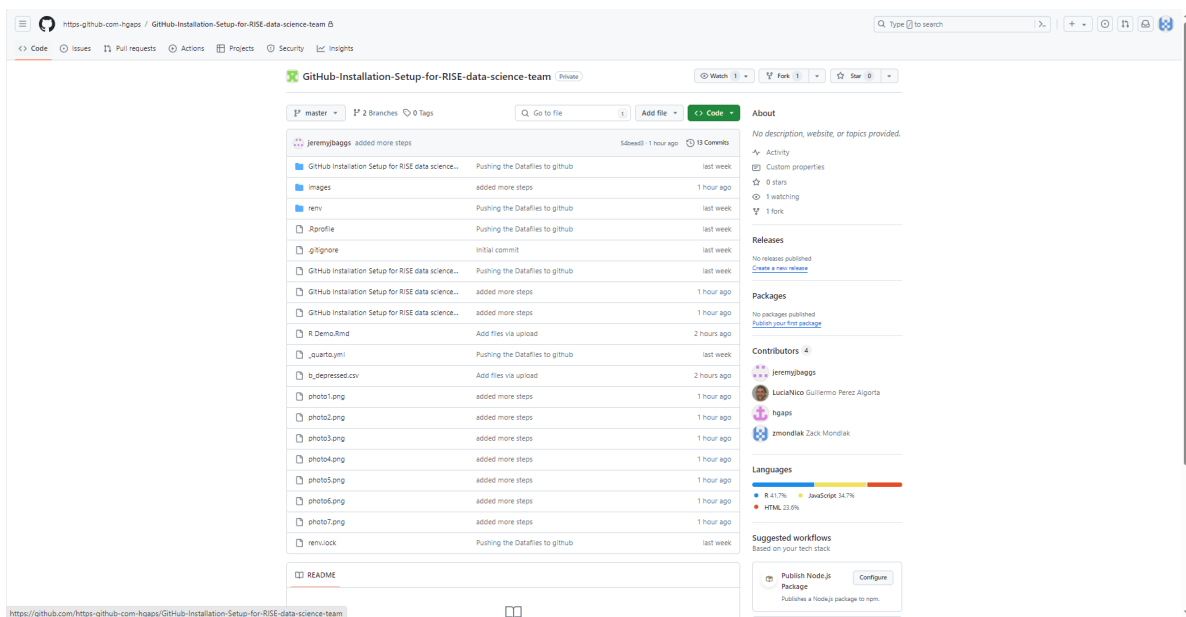
0.7 Step 5: Cloning a Git Repository in RStudio

Cloning a Git repository in RStudio allows you to create a local copy of a project from GitHub, enabling you to work on the code, make changes, and run analyses directly within the RStudio environment.

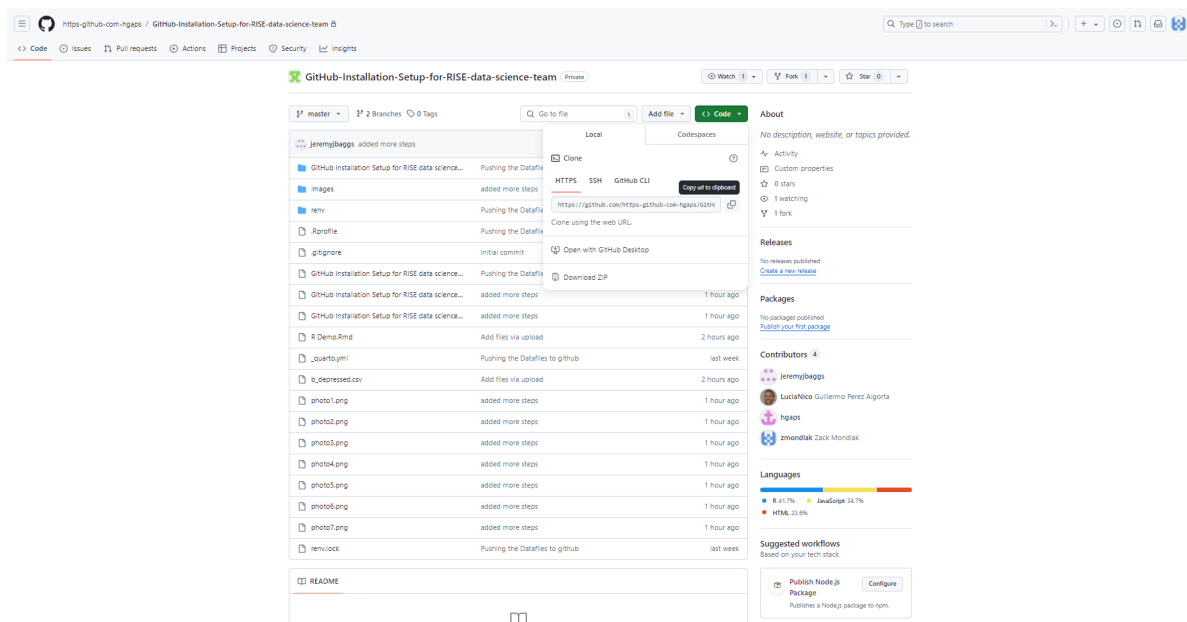
As mentioned, it facilitates seamless version control and collaboration by synchronizing your local changes with the remote repository.

First, let's access the GitHub repository supporting this tutorial.

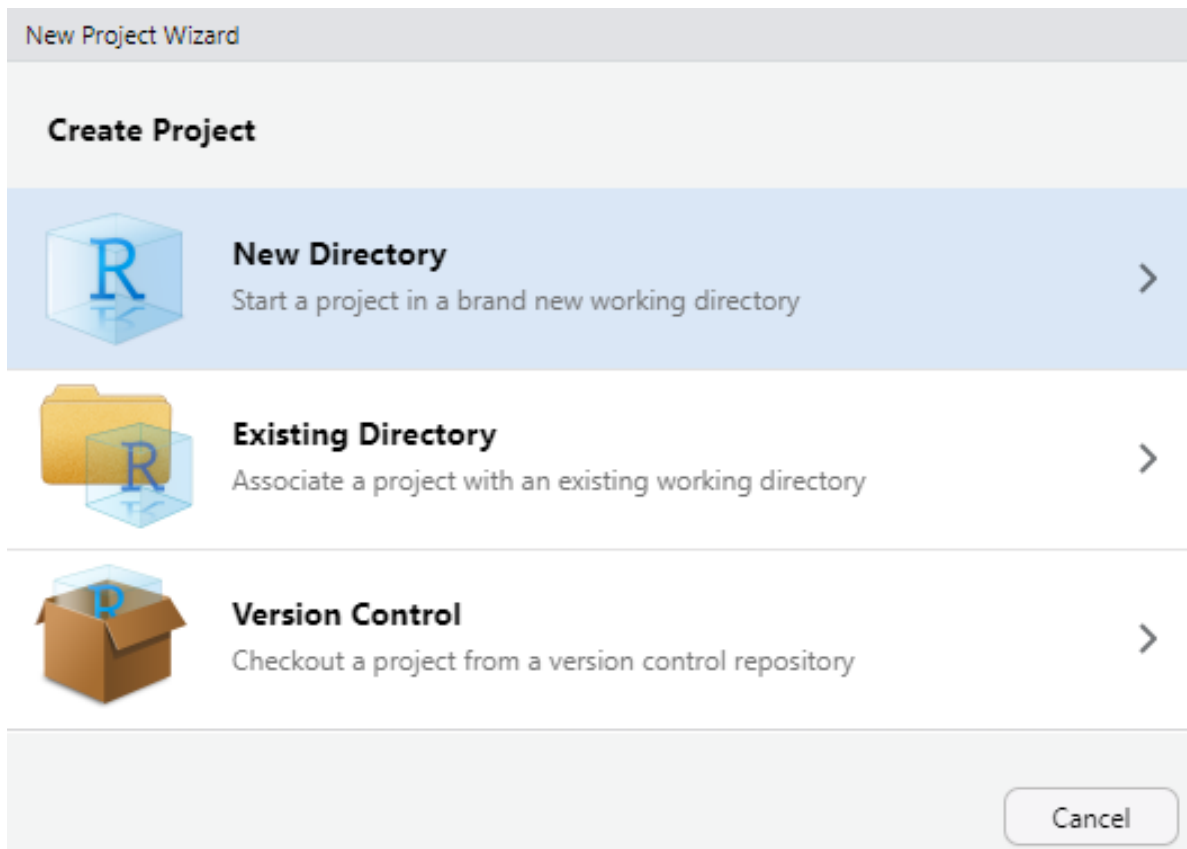
Click this link [add repo address...] and you should see this page.



Now, click on the green Code button. Then, copy the URL to the clipboard.



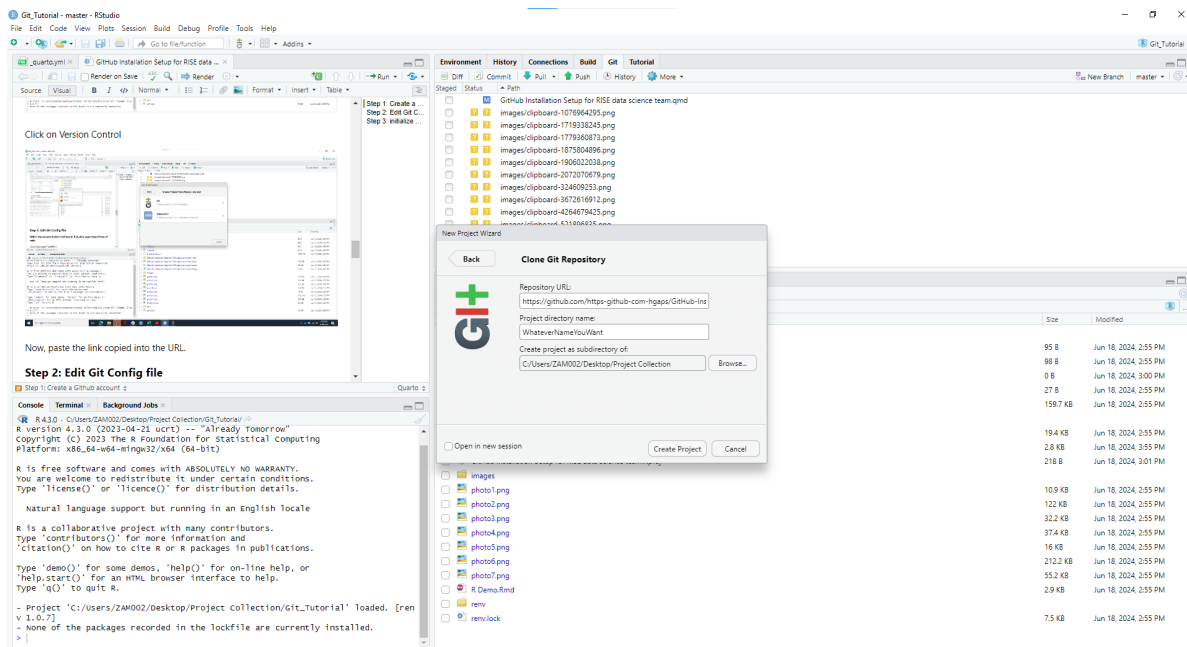
Open a new session in RStudio, and repeat the step mentioned above to arrive to this page:



But, now, instead of using New Directory, click on Version Control.

This will open a new window with two options, select Git.

Now, paste the link copied into the URL and click Create Project. Note: Be sure to save it somewhere you can find it easily!



Now, you have should have a copy of the remote repository, as a local project!!

Well done.